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New and Revised Data on Volumes of Brain Structures in Insectivores and Primates

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Abstract. More than 2,000 data on volumetric measurements of 42 structures in a variety of up to 76 species (28 insectivores, 21 prosimians, 27 simians) are given. All volumes measured in serial sections were converted to fresh volumes of a brain having a standard size within a given species. The data are available to all scientists for comparison and analysis. To allow critical evaluation, details on fixation and preparation, on determination of fresh brain weights and volumes of brain parts and on intraspecific variability are given.

Introduction

Comparative neuroanatomical investigations have demonstrated that the size of a given brain structure (i.e., the amount of brain tissue) is highly interrelated with (1) the size of the animal or species, and (2) the functional requirements of its habits [e.g., Stephan and Andy, 1969; Stephan and Pirlot, 1970]. The brain and its structures have to meet all external and internal requirements to allow the animals and species to survive. Higher and more complex behavioral patterns need larger and more differentiated cerebral structures. The two variables of a structure, i.e., size and differentiation, in general do not vary independently. Increased size is almost always accompanied by progressive differentiation and vice versa [e.g., Stephan and Andy, 1970; Stephan, 1975, for entorhinal cortex and olfactory bulb]. However, the *differentiation* can vary in many ways (e.g., in construction, arrangement and connection of the cells, units, layers, etc.), whereas for variations in *size,* there are only two possibilities: to become larger or smaller. Thus, an interspecific comparison of size may serve as a relatively simple first step in determining the functional importance (capacity) of a cerebral structure in a given species. We performed such interspecific comparisons using the allometric method. Differences in the size of brain structures were expressed by progression indices (enlargement factors) after due consideration of differences in body weight. Details on the method and results are given by Stephan and Andy [1969, 1977] and by Baron [1979].

Raw data on the volumes of the brain and of various brain parts in insectivores and primates have been published by Stephan et al. [1970], Frahm and Stephan [1976], Stephan and Andy [1977], and Baron [1977, 1979]. These data (1) made possible comparisons with similar data obtained in other laboratories; (2) allowed further analysis with appropriate mathematical methods (e.g., multivariate analysis); and (3) permitted reflections on possible interrelations with ethological and behavioral observations.

A renewed publication of data on volumetric measurements is desirable since: (1) data on many new specimens are available; (2) 23 of the new specimens belong to 13 new species; (3) within many old species [i.e., those already contained in Stephan et al., 1970], the inclusion of 61 new specimens results in alterations of the average values; (4) in several species 5 or more specimens could be measured, thus permitting more detailed insight into the variability of the volumes of brain structures; (5) the number of measured structures could be increased (most of these measurements were performed only on a part of the available serially sectioned brains); and (6) slight corrections of the standards of brain and body weights became necessary in some species due to new and extended information. For reasons of completeness, data from former studies which have not had to be modified are repeated here. For species in which more than 1 brain were measured, only the average volumes are given in the tables. Data on individual brains will be provided on request.

Since primates are thought to have had their phylogenetic origin in insectivore-like ancestors, data on insectivores are also included. The most primitive species of the extant insectivores probably have undergone little evolutionary change since their first appearance (permanent types). Hence, they can be expected to be comparatively similar to the early forerunners of primates and, therefore, to represent a good reference base for evaluating differing trends of brain parts during primate evolution.

Materials and Methods

Methods of Fixation and Preparation

Most of the animals were sacrificed with Nembutal, weighed and measured (head and body, tail, ear, hind foot). The brains were perfused in situ with Bouin's fluid after the blood was washed out with physiological saline. In smaller animals, the perfusion was performed through the heart, in larger ones through the carotids. After fixation with Bouin's fluid, brains have a hardened consistency and are uniformly yellow. By the uniform coloration they differ from other tissue; a factor favorable in the preparation. The brain is prepared by breaking small pieces of the skull away from the hardened brain. The operation begins at the foramen magnum and proceeds frontally on the base of the brain case. The vault of the cranium forms a bowl in which the brain rests during preparation. The upper jaw forms a kind of handle, with which the vault can be held. Subsequent to the preparation, the brain is weighed, its dimensions are measured, and then it is immersed in Bouin's fluid. After 4 days, it is preserved in alcohol (70%), which is changed several times during the 1st weeks.

With this method, even very small brains can be prepared uninjured, maintaining their natural proportions. In some specimens, however, perfusion is not possible. This is the case for human brains and for all animals which have died (e.g., in zoological gardens and animal houses) and cannot be perfused within the 1st h. Then the fresh (i.e., unfixed) brain must be prepared. Immersing larger fresh brains in Bouin's fluid is inappropriate, since this fixative hardens the surface and the infiltration is very slow. It is preferable to soak fresh brains in formalin.

Determination of Volumes

The brains were embedded in paraffin and sectioned serially (10, 15 or 20 μm thick). Cresyl-violet-stained sections taken at equal intervals (60–80 sections per brain) were used for volume estimation. The sections were projected directly on to photographic paper (18 × 24 cm), the borders delineated, and the components of each structure cut out and weighed. The photographic paper was previously weighed in order to determine the numbers of square millimeters per milligram, and care was taken to compensate for changes in paper weight due to varying temperature and humidity. Distances between the histologic sections were calculated, and the volumes for each division were derived by applying the formula:

$$V = \frac{AP \cdot WS \cdot D}{M^2},$$

where V = volume in mm^3; AP = average area of the photographic paper in mm^2/mg; WS = weight of the photographic paper of the cut-out structure in mg; D = distance of measured sections in mm; M^2 = square of linear magnification.

The sum of the volumes of all measured structures (= total brain) obtained in this manner is markedly less than the volume of the fresh brain. This difference is due to shrinkage resulting from fixation and embedding. The degree of shrinkage is different for each brain, even if these procedures are conducted in the same way. To obtain comparable values, it is necessary to correct the figures to the volume of the *fresh* brain. This value can be obtained by dividing the weight of the fresh brain by the specific brain weight. The specific gravity of fresh brains was found to be close to 1.036 g/cm³ [Stephan, 1960]. Thus,

$$\text{volume of fresh brain} = \frac{\text{weight of fresh brain}}{1.036}$$

To obtain the conversion factor for an individual brain (C_{ind}), this volume must be divided by the serial section volume:

$$C_{ind} = \frac{\text{Volume of fresh brain}}{\text{serial section volume}} .$$

The size of this conversion factor is a function of the specific type of fixation. The following means were found:

Fixation fluid	n	Factor of conversion		Volume loss in % of fresh brain	
		mean	range	mean	range
Bouin's fluid	315[1]	1.94 ± 8.92%	1.65–2.57	48.3	39.6–61.1
Formol-alcohol	5	1.54 ± 7.65%	1.41–1.73	35.0	29.0–42.0
Formol 10%	21	2.40 ± 9.35%	2.02–2.86	58.4	50.6–65.1

[1] Chiroptera included.

Because of these large differences, it is inappropriate to compare uncorrected figures from material with unknown or different types of fixation. Even if the same fixation and embedding procedures are applied to a series of brains, conversions remain necessary because of the wide range of volume loss to be expected as indicated in the last column of the above table.

The conversion factor calculated for the *total* brain is then used to correct all the *parts*. This may not be quite correct, since various parts of the brain may have somewhat different degrees of shrinkage. However, such differences are difficult to ascertain and since they are expected to be small, they were not taken into account.

Introduction of Standards. To obtain volumes that are as representative as possible for a given *species,* we have performed a second conversion (C_{sta}) that takes into account the difference between the individual weight of the brain treated here and the standard brain weight of each species as given by Bauchot and Stephan [1966, 1969]. Such differences are due to intraspecific variations in the brain weights.

We introduced these standards since, in general, our information about average brain size within a given species is much more extensive than our knowledge about the average volume size of the various brain structures. These were the reasons: (1) even when there is ample brain material from a given species, it would be cumbersome and, to some extent, useless to utilize all this material for comparative studies merely to have a better base on which to determine the average. The number of brains we have sectioned serially is limited, since such work is very time-consuming and expensive. Furthermore, for our purposes, consideration of as many *species* as possible is more informative than the investigation of many *specimens* of single species; (2) by forming standards, serial-sectioned brains for which the brain weight is not known can be used for interspecific volume comparisons; (3) standards generally permit the inclusion of data taken from the literature that otherwise would have to be rejected.

The last two points are of special importance in species that are valuable and/or rare and, more especially, that are necessary to conserve. The most extensive information about total brain size generally can be incorporated into the standard volumes of the various brain parts by using

$$C_{sta} = \frac{\text{weight of standard fresh brain}}{\text{weight of individual fresh brain}}.$$

This second conversion (C_{sta}) again was determined by comparing *total* brain, and was utilized to correct all the *parts* of the brain. This is possible since, from our experience, extremely small or extremely large brains within a given species generally have the same composition. Intraspecific comparisons of small and large brains have not shown any directed changes in the composition of the brain [e.g., Kruska and Stephan, 1973].

We believe that by applying standard brain weights for each species, we are able to obtain from a few or even from only one specimen, values that show a good approximation to the unknown mean.

In addition to *standard brain weights*, we used *standard body weights* to determine standard indices using the allometric method (see 'Introduction'). The body weight standards are the most probable values as derived from all available material after exclusion of juvenile, thin and sick, as well as excessively fat animals, and after conversion of the body weights of pregnant females. Thus, they are sometimes notably different from the arithmetic means taken from all known data. Due to the addition of new data, slight alterations of the standard values given by Bauchot and Stephan [1966, 1969] and Stephan et al. [1970] became necessary for a few species.

Results

Determination of Fresh Brain Weights

We have pointed out that comparable values can hardly be obtained without correcting the figures in relation to fresh brain. However, most of our brains are *not* prepared fresh, but after fixation in situ by perfusion with Bouin's fluid. The reason for this is that in very small animals (e.g., shrews) generally only a pulp of brain tissue is obtainable from fresh preparations and in all kinds of brains, the natural proportions are distinctly better preserved when fixed in situ. In situ fixation makes it possible to prepare uninjured and complete brains of all kinds and sizes, and thus meets the requirements for exact measurements.

The question arises as to whether the method used here preserves the fresh brain weight at least during the time of fixation and preparation, or if the brain weights taken after perfusion with Bouin's fluid need a correction to arrive at fresh brain weights. According to investigations in guinea pigs (fig. 1), the weight of a brain fixed in Bouin's fluid corresponds to the weight of the fresh brain when prepared within the first (about 5) hours. Then it slowly decreases [Stephan, 1960].

These results were confirmed by new studies in rats: no differences were found between the weights of the brains without perfusion (fresh brain weight) and with perfusion by Bouin's fluid, when the brains were prepared and weighed immediately after perfusion. The average brain weight is the same in both groups. All values are mixed thoroughly when plotted against body weight, and there is not even a slight tendency towards enlargement or reduction.

Brain Weight Alterations in Bouin's Fluid. It is of interest to know what happens when animals or heads are perfused and the brains cannot be prepared immediately (e.g., when animals are collected in foreign countries by scientists having no experience in brain preparations). We have several such brains in our collection, some of which are of rare species. Do the brain weights react (1) like those of brains freshly immersed in Bouin's fluid (triangles in fig. 2), or (2) like those of brains immediately prepared and weighed after perfusion with Bouin's fluid (squares in fig. 2), or (3) in other ways?

(1) Brains *freshly* immersed in Bouin's fluid became somewhat larger the 1st day (up to 2%), and then slowly decreased in weight, reaching the original weight and dropping below it the 2nd day. Leibnitz [1967], however, found no initial enlargement in rat brains.

(2) The weight of *perfused* brains began to decline immediately after immersion (squares in fig. 2). These results agree well with those of Bauchot [1967]. Weight loss in the perfused brains proceeded at a lower rate than in nonperfused brains (triangles in fig. 2). Curves 1 and 2 in figure 2 cross at about the 16th day when the weight decrease was about 9%. After 3 months, the average decrease in the perfused brain was about 16%, that of the nonperfused brains, about 20%.

(3) For brains perfused with Bouin's fluid but not prepared immediately, Bauchot [1967] found a different behavior: no change of brain weight in

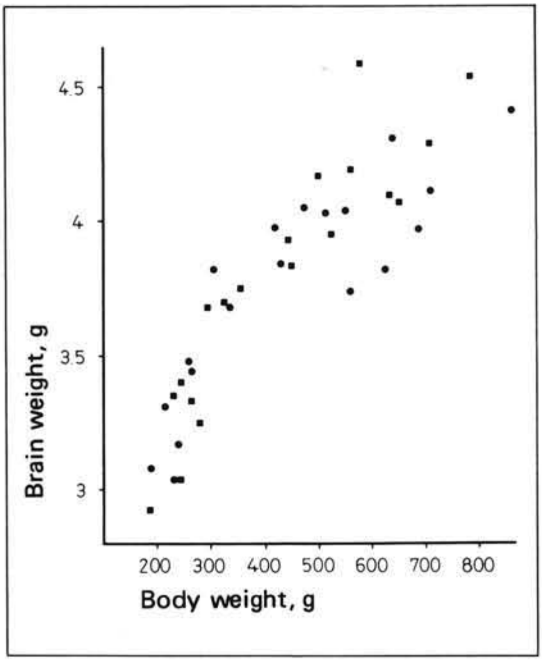

Fig. 1. Brain weights of guinea pigs *(Cavia cobaya)* after preparing the brains fresh (●, n = 19) and after in situ perfusion with Bouin's fluid when preparing and weighing the brains immediately after perfusion (■, n = 19). There are no recognizable differences in brain weights between the two samples.

at least the first 16 days. This is in contrast to our own experiences with Mexican bats which could not be prepared before 3 months after perfusion. Their brain weights indicated a clear reduction when compared with the average brain weight taken from specimens of the same species which were prepared immediately after perfusion.

To get some insight into the possible alteration in brain weight when preparation of the brain is delayed, we perfused 8 male and 8 female rats (group I), immersed the heads in Bouin's fluid and prepared and weighed one pair of brains at a time 1, 2, 4, 7, 11, 16, 22 and 29 days after fixation. In group II, 24 rat brains were prepared fresh or immediately after perfusion with Bouin's fluid. For this group, the interrelation between body and brain weight was determined by calculating the regression line. The line specifies which brain weight is on the average correlated with which given body weight.

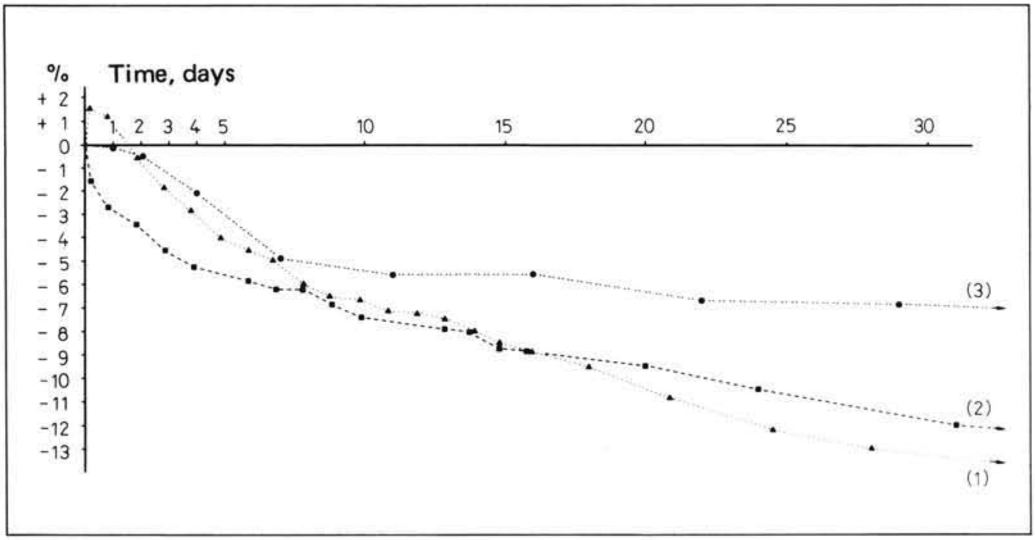

Fig. 2. Percentage deviation from reference brain weights in rats plotted against time. (1) The fresh brain weights (reference) of 12 rats were taken immediately after having sacrificed the animals and prepared the brains. The brains were then immersed in Bouin's fluid and weighed at various intervals. ▲ = The average deviation of these weights from the individual reference brain weights. (2) 12 rats were perfused with Bouin's fluid and the brains prepared and weighed immediately. The brain weights obtained in this manner (reference) do not differ from fresh brain weights (see also figure 1, guinea pigs). The brains were then immersed in Bouin's fluid and weighed at different intervals. ■ = The average deviation of these weights from the individual reference brain weights. (3) 16 rats (8 males and 8 females) were perfused with Bouin's fluid and at intervals of 1, 2, 4, 7, 11, 16, 22 and 29 days, the brains of one pair were prepared and weighed. The individual brain weights obtained after the different time periods were related to the individual reference brain weights, as determined on the basis of the individual body weights from the brain-weight/body-weight regression line constructed for the 24 rats referred to under (1) and (2) (see also text). The individual differences between the measured brain weights and the calculated reference weights were averaged for each pair (●).

The brain weights of group I animals were related to the brain weights determined from the group II regression line for the individual body weights of the group I animals. The results are shown in figure 2 (curve 3). Because of the method used, a smooth curve cannot be expected, but as the figure shows, there was a clear tendency for brain weight to decline. Although the decrease was less, it was observable during the first days. After 16 days, there was an average decrease of about 5.5%, after 1 month nearly 7%. This reduction is clearly less than that found both in fresh and in perfused and immediately prepared brains, when both are subsequently immersed in Bouin's fluid (fig. 2, curves 1 and 2). Figure 2 (curve 3) may serve to correct the weight of brains which could not be prepared immediately after perfusion. The approach is, of course, rough, but may give approximate values.

Survey of Structures and Complexes

Volumes of structures and complexes are given for:

(A) 76 species (28 insectivores: table I; 21 prosimians: table II; 27 simians: table III) for the ventricles (3), meninges, nerves, etc. (4), medulla oblongata (6), cerebellum (7), mesencephalon (8), and diencephalon (9);

(B) 76 species (28 insectivores: table IV; 21 prosimians: table V; 27 simians: table VI) for telencephalon (10), bulbus olfactorius (11), bulbus olfactorius accessorius (12), lobus piriformis (13), septum (14), striatum (15), schizocortex (16), hippocampus (17), and neocortex (18);

(C) 37 species (10 insectivores: table VII; 12 prosimians and 15 simians: table VIII) for epithalamus (19), thalamus (20), hypothalamus (21), subthalamus (22), pallidum (23), nucleus subthalamicus Luysi (24), diencephalic part of capsula interna (25), and tractus opticus (26);

(D) 40 species (20 prosimians and 20 simians: table IX) for area striata (27), and corpus geniculatum laterale (28);

(E) 28 species (10 insectivores: table X; 10 prosimians and 8 simians: table XI) for palaeocortex (29), amygdala (30) with centromedial and cortico-basolateral nuclear groups (31–34);

(F) 37 species (10 insectivores: table XII; 12 prosimians and 15 simians: table XIII) for the vestibular complex (35) with nucleus vestibularis superior (36), lateralis (37), medialis (38), and descendens (39);

(G) 67 species (26 insectivores: table XIV; 20 prosimians: table XV; 21 simians: table XVI) for nucleus septalis triangularis (40), corpus subfornicale (41), nucleus habenularis medialis (42), corpus pineale (43), and corpus subcommissurale (44).

Variability

All volume data of the tables are given with at least three real numbers, e.g., 0.000111, 0.00111, 0.0111, 0.111, 1.11, 11.1, 111, 1,111, or 11,111 etc. In cases of very large figures, the last places are, of course, fairly unimportant since a variability of up to 10% may be expected. In most species (48 of 76), the 15 main structures and/or complexes given in tables I–VI were measured in more than 1 brain and the standard error of the mean was

Table I. Data on body weights (g), brain weights (mg), and volumes of total brain and fundamental brain parts (mm³) in insectivores

Code number	Species	Number of individuals	Body weight	Brain weight	Ventricles	Meninges, hypophysis, nerves, etc.	Total brain (net volume)	Medulla oblongata	Cerebellum	Mesencephalon	Diencephalon
			1	2	3	4	5 (= 6–10)	6	7	8	9
0001	*Solenodon paradoxus*	2	900	4,670	56.4	169	4,282	466	753	206	269
0002	*Tenrec ecaudatus*	3	832	2,570	65.3	101	2,315	313	309	132	153
0003	*Setifer setosus*	2	248	1,510	21.3	32.3	1,404	183	142	75.7	99.7
0004	*Hemicentetes semispin.*	1	110	830	15.5	29.0	757	108	117	44.8	48.5
0005	*Echinops telfairi*	2	87.5	620	8.34	23.9	566	89.9	57.9	43.8	40.1
0006	*Oryzorictes talpoides*	1	44.2	580	4.83	17.3	538	61.0	81.3	26.6	36.9
0007	*Microgale cowani*	1	15.2	420	7.12	6.40	392	45.3	49.5	24.0	25.4
0008	*Limnogale mergulus*	1	92.0	1,150	12.6	51.0	1,046	204	176	67.4	81.0
0009	*Nesogale dobsoni*	1	32.6	560	5.31	21.9	513	59.0	57.3	37.9	32.8
0010	*Nesogale talazaci*	1	50.4	790	6.02	16.0	741	85.3	92.3	47.2	43.8
0014	*Micropotamogale lamottei*	3	64.2	800	11.9	17.2	743	135	82.7	58.6	62.8
0015	*Potamogale velox*	2	660	4,160	16.6	121	3,878	756	605	254	298
0016	*Chlorotalpa stuhlmanni*	2	39.8	740	7.00	13.8	693	68.3	74.7	42.0	51.8
0017	*Chrysochloris asiatica*	1	49.0	700	8.12	10.7	657	58.5	72.6	43.2	52.8
0021	*Aethechinus algirus*	3	790	3,200	121	109	2,859	336	358	153	246
0022	*Erinaceus europaeus*	5	860	3,350	105	78.8	3,050	333	394	151	241
0023	*Hemiechinus auritus*	3	250	1,900	82.5	41.2	1,710	187	198	110	142
0031	*Elephantulus fuscipes*	2	57.0	1,330	18.1	32.9	1,233	120	183	117	112

0033	*Rhynchocyon stuhlmanni*	2	490	6,100	97.2	111	5,680	507	798	523	524
0041	*Sorex minutus*	1	5.3	110	0.46	2.47	103	17.0	11.0	5.91	8.69
0042	*Sorex araneus*	5	10.3	200	1.17	3.96	188	27.1	20.4	10.8	13.7
0043	*Neomys fodiens*	2	15.2	320	1.63	8.13	299	42.2	38.9	17.3	25.2
0059	*Crocidura occidentalis*	2	28.0	440	2.67	14.0	408	59.8	51.3	27.7	28.4
0060	*Crocidura russula*	2	11.0	190	1.67	3.85	178	23.6	21.8	11.7	15.0
0063	*Suncus murinus*	1	35.5	380	4.60	7.80	354	50.5	44.8	24.2	26.0
0081	*Talpa europaea*	2	76.0	1,020	10.9	20.6	953	95.4	174	49.8	80.4
0083	*Desmana moschata*	1	440	4,000	138	104	3,620	402	514	219	344
0084	*Galemys pyrenaicus*	2	57.5	1,330	27.5	25.9	1,230	147	184	62.2	105

1, 2: Body weight and brain weight standards were taken from Bauchot and Stephan [1966, 1969]. Alterations became necessary for a few species, due to extension of material and/or additional information from literature.
3, 4: Ventricles, parts of brain nerves, and remnants of meninges [and hypophysis, see diencephalon, (9)] are regarded not to be pure brain tissues. They are included in the measurements, since they are included also when taking the individual brain weight.
5: Net volume of pure tissue is total brain volume without ventricles (3), nerves, meninges, etc. (4).
6: Since it is almost impossible to separate distinctly the substantia reticularis of the medulla oblongata from that of the mesencephalon, the latter is included into the medulla oblongata.
7: Included are brachium and nuclei pontis.
8: Without substantia reticularis, see 6.
9: Included is the pallidum (globus pallidus). In order to get comparable values, the hypophysis is excluded from the diencephalon (and included in 4) since it is not preserved in all brains (e.g., human), due to difficulties in preparation.

Table II. Data on body weights (g), brain weights (mg), and volumes of total brain and fundamental brain parts (mm^3) in prosimians; for comments on structures see table I

Code number	Species	Number of individuals	Body weight	Brain weight	Ventricles	Meninges, hypophysis, nerves, etc.	Total brain (net volume)	Medulla oblongata	Cerebellum	Mesencephalon	Diencephalon
			1	2	3	4	5 (= 6–10)	6	7	8	9
0102	*Tupaia glis*	20	170	3,200	38.2	51.4	2,999	219	376	256	275
0104	*Tupaia minor*	2	70	2,580	22.2	37.8	2,430	173	308	249	251
0106	*Urogale everetti*	1	275	4,280	38.9	95.8	3,997	355	513	307	394
0111	*Cheirogaleus major*	2	450	6,800	83.4	108	6,373	382	947	241	489
0112	*Cheirogaleus medius*	2	177	3,140	24.6	45.2	2,961	202	437	145	235
0114	*Microcebus murinus*	6	54	1,780	11.3	26.6	1,680	99.1	234	85.2	134
0117	*Lepilemur ruficaudatus*	3	915	7,600	76.5	84.1	7,175	449	1,165	260	593
0118	*Lemur fulvus*	3	1,400	23,300	194	190	22,106	909	3,328	590	1,713
0124	*Lemur variegatus*	1	3,000	31,500	299	393	29,713	1,420	4,286	1,018	2,529
0130	*Avahi l. laniger*	2	1,285	10,490	172	156	9,798	553	1,489	343	862
0131	*Avahi l. occidentalis*	2	860	9,670	73.7	137	9,124	508	1,383	345	861
0132	*Propithecus verreauxi*	2	3,480	26,700	231	347	25,194	1,223	3,957	780	2,195
0133	*Indri indri*	2	6,250	38,300	330	354	36,285	1,342	5,504	978	2,911
0136	*Daubentonia madagasc.*	1	2,800	45,150	392	578	42,611	1,517	6,461	897	3,539
0137	*Loris tardigradus*	1	322	6,600	51.6	50.4	6,269	233	728	220	538
0138	*Nycticebus coucang*	2	800	12,500	142	169	11,755	528	1,310	345	1,077
0139	*Perodicticus potto*	2	1,150	14,000	127	174	13,212	680	1,699	391	1,024
0141	*Galago crassicaudatus*	2	850	10,300	147	128	9,668	540	1,414	384	785
0142	*Galago demidovii*	2	81	3,380	30.4	28.8	3,203	169	413	135	246
0143	*Galago senegalensis*	1	186	4,800	40.2	81.4	4,512	254	672	205	382
0147	*Tarsius* sp.	2	125	3,600	24.1	58.0	3,393	207	428	197	303

Table III. Data on body weights (g), brain weights (mg), and volumes of total brain and fundamental brain parts (mm^3) in simians; for comments on structures see table I

Code number	Species	Number of individuals	Body weight	Brain weight	Ventricles	Meninges, hypophysis, nerves, etc.	Total brain (net volume)	Medulla oblongata	Cerebellum	Mesencephalon	Diencephalon
			1	2	3	4	5 (= 6–10)	6	7	8	9
0151	*Callithrix jacchus*	4	280	7,600	52.0	42.9	7,241	318	757	295	554
0156	*Cebuella pygmaea*	2	140	4,500	24.5	16.9	4,302	201	468	192	342
0164	*Saguinus oedipus*	3	380	10,000	62.6	52.8	9,537	413	984	333	754
0165	*Saguinus tamarin*	2	340	10,300	251	122	9,569	428	1,061	332	692
0166	*Callimico goeldii*	1	480	11,000	47.9	59.5	10,510	460	1,240	340	739
0171	*Aotus trivirgatus*	5	830	17,100	105	206	16,195	686	1,873	409	1,098
0172	*Callicebus moloch*	2	900	19,000	279	242	17,944	787	1,622	530	1,375
0174	*Pithecia monacha*	2	1,500	35,000	631	286	32,867	1,009	3,908	745	2,285
0179	*Alouatta* sp.	2	6,400	52,000	616	568	49,009	1,593	5,699	986	3,344
0184	*Ateles geoffroyi*	1	8,000	108,000	1,389	1,824	101,034	1,834	12,438	1,482	5,334
0186	*Lagothrix lagotricha*	3	5,200	101,000	1,090	897	95,503	2,110	11,268	1,582	5,721
0191	*Cebus* sp.	2	3,100	71,000	729	864	66,939	1,738	7,871	1,221	3,996
0192	*Saimiri sciureus*	1	660	24,000	299	295	22,572	722	2,260	526	1,430
0202	*Macaca mulatta*	1	7,800	93,000	834	1,038	87,896	1,992	8,965	1,380	4,480
0209	*Cercocebus albigena*	1	7,900	104,000	742	2,041	97,603	2,708	10,726	1,770	5,351
0212	*Papio anubis*	2	25,000	201,000	1,081	1,977	190,957	5,297	18,683	2,711	9,280
0227	*Cercopithecus mitis*	1	6,300	75,000	467	1,363	70,564	1,999	6,758	1,354	4,176
0231	*Cercopithecus ascan.*	1	3,400	67,000	240	927	63,505	1,632	5,828	1,162	3,605
0232	*Cercopithecus talap.*	2	1,200	40,000	262	572	37,776	1,035	3,374	826	2,375
0234	*Erythrocebus patas*	2	7,800	108,000	561	519	103,167	2,616	8,738	1,621	5,423
0242	*Pygathrix nemaeus*	1	7,500	77,000	911	883	72,530	2,206	8,063	1,345	4,727
0243	*Nasalis larvatus*	1	14,000	97,000	591	241	92,797	2,945	12,113	1,556	5,310
0246	*Colobus badius*	2	7,000	78,000	455	1,016	73,818	2,007	8,648	1,333	3,945
0271	*Hylobates lar*	1	5,700	102,000	555	395	97,505	2,251	12,078	1,459	5,716
0277	*Pan troglodytes*	1	46,000	405,000	1,899	6,924	382,103	5,817	43,663	3,739	15,392
0279	*Gorilla gorilla*	1	105,000	500,000	3,608	8,659	470,359	7,509	69,249	4,352	19,370
0280	*Homo sapiens sapiens*	1	65,000	1,330,000	18,732	13,205	1,251,847	9,622	137,421	8,087	33,319

Table IV: Volumes of the telencephalon and of its components (mm^3) in insectivores

Code number	Species	Number of individuals	Telencephalon	Bulbus olfactorius	Bulbus olfactorius accessorius	Lobus piriformis	Septum	Striatum	Schizocortex	Hippocampus	Neocortex
			10 (= 11–18)	11	12	13	14	15	16	17	18
0001	*Solenodon paradoxus*	2	2,486	465	0.218	689	65.4	255	156	296	661
0002	*Tenrec ecaudatus*	3	1,407	314	0.715	464	36.9	98.1	76.5	146	271
0003	*Setifer setosus*	2	903	209	0.534	336	26.0	56.6	41.9	99.2	134
0004	*Hemicentetes semispin.*	1	438	93.6	0.196	148	12.8	29.8	17.4	63.7	73.4
0005	*Echinops telfairi*	2	335	71.2	0.249	116	11.5	24.1	13.1	46.4	51.5
0006	*Oryzorictes talpoides*	1	332	46.9	0.120	88.0	10.0	32.5	23.5	64.3	66.5
0007	*Microgale cowani*	1	248	35.9	0.162	79.0	6.08	19.4	13.5	36.4	57.2
0008	*Limnogale mergulus*	1	518	42.9	0.323	87.4	19.5	57.9	36.4	78.2	196
0009	*Nesogale dobsoni*	1	326	53.6	0.178	93.5	9.58	23.1	16.3	67.0	62.9
0010	*Nesogale talazaci*	1	472	74.3	0.369	139	11.2	31.2	20.7	98.9	96.2
0014	*Micropotamogale lamottei*	3	404	34.0	0.0605	67.3	15.1	32.6	32.9	77.7	144
0015	*Potamogale velox*	2	1,964	86.0	2.31	194	51.4	134	145	298	1,054
0016	*Chlorotalpa stuhlmanni*	2	457	60.6	0.204	131	11.3	43.5	21.4	59.1	129
0017	*Chrysochloris asiatica*	1	430	58.5	0.132	115	12.1	44.3	20.9	66.3	113
0021	*Aethechinus algirus*	3	1,766	285	1.93	459	42.9	140	109	227	500
0022	*Erinaceus europaeus*	5	1,931	332	1.94	505	51.7	147	134	237	522
0023	*Hemiechinus auritus*	3	1,073	175	1.35	279	28.4	81.1	68.7	130	309
0031	*Elephantulus fuscipes*	2	700	63.3	0.662	120	16.8	44.5	46.8	173	236
0033	*Rhynchocyon stuhlmanni*	2	3,328	426	0.666	705	81.5	222	220	582	1,089

0041	*Sorex minutus*	1	60.7	8.65	0.0202	17.2	2.02	6.02	3.62	10.5	12.7
0042	*Sorex araneus*	5	116	15.7	0.0776	32.7	3.20	10.5	6.83	20.6	26.4
0043	*Neomys fodiens*	2	175	15.8	0.101	36.0	6.42	18.7	9.00	32.9	56.6
0059	*Crocidura occidentalis*	2	241	37.6	0.221	68.4	6.94	16.8	13.3	38.5	59.1
0060	*Crocidura russula*	2	106	16.3	0.0431	30.2	3.64	8.17	6.05	15.9	25.5
0063	*Suncus murinus*	1	209	34.2	0.182	56.2	5.50	14.8	12.6	31.2	54.1
0081	*Talpa europaea*	2	554	59.6	0.291	112	15.0	60.2	39.6	86.8	181
0083	*Desmana moschata*	1	2,140	141	0.218	320	42.6	222	163	267	983
0084	*Galemys pyrenaicus*	2	733	39.0	0.206	107	20.1	74.4	44.9	109	339

13: Included are palaeocortex (29) and amygdala (30); excluded is the schizocortex (16).

14: Included are septum pellucidum, septum verum, Broca's diagonal band in its whole extent, and the bed nuclei of the anterior commissure and of the stria terminalis.

15: Included are caudate nucleus, putamen, nucleus accumbens, and the parts of the capsula interna running through the striatum. In low insectivores, these parts of the internal capsule are strongly dispersed and difficult to separate from the striatum. Excluded is the pallidum (9).

16: Included are entorhinal, perirhinal, presubicular and parasubicular cortices and the underlying white matter. These cortices are characterized by the presence of one or several, almost cell-free layers or sublayers.

17: Included are retro- or postcommissural hippocampus as well as supra- and precommissural hippocampus.

18: Included are the underlying white matter and the corpus callosum. From the transitional cortices are included insular, subgenual, cingular and retrosplenial regions.

Table V. Volumes of the telencephalon and of its components (mm³) in prosimians; for comments on structures see table IV

Code number	Species	Number of individuals	Telencephalon	Bulbus olfactorius	Bulbus olfactorius accessorius	Lobus piriformis	Septum	Striatum	Schizocortex	Hippocampus	Neocortex
			10 (= 11–18)	11	12	13	14	15	16	17	18
0102	*Tupaia glis*	20	1,873	131.5	2.41	269	27.5	140	114	152	1,036
0104	*Tupaia minor*	2	1,449	94.3	2.28	205	26.1	124	91.9	132	774
0106	*Urogale everetti*	1	2,428	184	2.10	362	45.1	199	124	173	1,338
0111	*Cheirogaleus major*	2	4,314	158	4.04	332	58.9	327	149	347	2,938
0112	*Cheirogaleus medius*	2	1,941	102	2.66	198	32.5	145	65.3	174	1,221
0114	*Microcebus murinus*	6	1,129	43.0	1.62	103	15.3	85.7	40.1	100	740
0117	*Lepilemur ruficaudatus*	3	4,708	113	2.72	302	56.7	366	194	392	3,282
0118	*Lemur fulvus*	3	15,566	207	3.21	641	119	1,215	422	752	12,207
0124	*Lemur variegatus*	1	20,461	369	5.27	985	206	1,591	608	1,404	15,293
0130	*Avahi l. laniger*	2	6,550	89.2	3.01	284	69.7	524	241	526	4,813
0131	*Avahi l. occidentalis*	2	6,026	70.1	2.55	265	69.3	478	222	476	4,443
0132	*Propithecus verreauxi*	2	17,040	147	3.40	664	159	1,411	443	1,043	13,170
0133	*Indri indri*	2	25,551	168	2.32	838	205	1,855	848	1,520	20,114
0136	*Daubentonia madagasc.*	1	30,196	685	8.31	1,538	254	2,765	1,044	1,776	22,127
0137	*Loris tardigradus*	1	4,551	85.8	2.30	256	41.1	351	99.7	191	3,524
0138	*Nycticebus coucang*	2	8,495	159	4.28	510	92.0	760	212	566	6,192
0139	*Perodicticus potto*	2	9,418	310	2.74	661	114	712	328	607	6,683
0141	*Galago crassicaudatus*	2	6,545	166	2.52	386	70.0	556	181	461	4,723
0142	*Galago demidovii*	2	2,240	83.1	1.26	181	25.7	169	59.9	152	1,568
0143	*Galago senegalensis*	1	2,997	79.2	2.58	163	35.7	210	107	261	2,139
0147	*Tarsius* sp.	2	2,258	18.8	0.820	92.7	23.3	133	69.0	153	1,768

Table VI. Volumes of the telencephalon and of its components (mm³) in simians; for comments on structures see table IV

Code number	Species	Number of individuals	Telencephalon	Bulbus olfactorius	Bulbus olfactorius accessorius	Lobus piriformis	Septum	Striatum	Schizocortex	Hippocampus	Neocortex
			10 (= 11–18)	11	12	13	14	15	16	17	18
0151	*Callithrix jacchus*	4	5,318	22.8	0.326	191	49.8	372	90.0	221	4,371
0156	*Cebuella pygmaea*	2	3,099	12.3	0.386	133	29.2	174	81.8	133	2,535
0164	*Saguinus oedipus*	3	7,052	19.1	0.406	256	60.5	453	107	262	5,894
0165	*Saguinus tamarin*	2	7,055	17.3	0.290	243	40.9	471	120	280	5,883
0166	*Callimico goeldii*	1	7,733	27.0	0.512	253	63.5	493	138	281	6,476
0171	*Aotus trivirgatus*	5	12,128	56.0	0.222	394	83.8	862	243	539	9,950
0172	*Callicebus moloch*	2	13,465	19.2	0.325	454	86.4	920	234	588	11,163
0174	*Pithecia monacha*	2	24,920	34.8	0.846	675	140	1,918	289	834	21,028
0179	*Alouatta* sp.	2	37,388	41.4	0.595	827	200	2,829	510	1,320	31,660
0184	*Ateles geoffroyi*	1	79,946	90.4	2.19	1,625	324	4,950	732	1,366	70,856
0186	*Lagothrix lagotricha*	3	74,822	73.4	1.43	1,395	266	4,947	680	1,586	65,873
0191	*Cebus* sp.	2	52,113	39.9	0.667	931	174	3,258	390	890	46,429
0192	*Saimiri sciureus*	1	17,635	26.8	0.985	413	90.6	1,042	168	352	15,541
0202	*Macaca mulatta*	1	71,080	84.3	0	1,220	271	4,032	639	1,353	63,482
0209	*Cercocebus albigena*	1	77,049	121	0	1,639	294	4,146	630	1,485	68,733
0212	*Papio anubis*	2	154,987	287	0	2,111	559	7,182	1,309	3,398	140,142
0227	*Cercopithecus mitis*	1	56,277	117	0	1,265	246	2,733	617	1,366	49,933
0231	*Cercopithecus ascan.*	1	51,279	99.5	0	1,052	251	2,827	694	1,189	45,166
0232	*Cercopithecus talap.*	2	30,166	27.8	0	706	133	1,908	259	705	26,427
0234	*Erythrocebus patas*	2	84,770	51.8	0	1,339	330	3,624	693	1,591	77,141
0242	*Pygathrix nemaeus*	1	56,189	10.7	0	942	290	3,166	723	2,295	48,763
0243	*Nasalis larvatus*	1	70,873	30.3	0	1,268	333	3,735	855	1,966	62,685
0246	*Colobus badius*	2	57,885	51.3	0	937	288	3,217	814	1,671	50,906
0271	*Hylobates lar*	1	76,001	43.9	0	1,264	302	4,784	1,136	2,673	65,800
0277	*Pan troglodytes*	1	313,493	257	0	2,750	851	12,246	2,018	3,779	291,592
0279	*Gorilla gorilla*	1	369,878	316	0	4,871	1,173	14,567	2,729	4,781	341,444
0280	*Homo sapiens sapiens*	1	1,063,399	114	0	9,032	2,610	28,689	6,142	10,287	1,006,525

0 = Structure is not present in the species under consideration.

Table VII. Volumes of components of the diencephalon (mm³) in insectivores

Code number	Species	Number of individuals	Epithalamus	Thalamus	Hypothalamus	Subthalamus	Pallidum	Nucleus subthalamicus	Capsula interna	Tractus opticus
			19	20	21	22 (= 23 + 24)	23	24	25	26
0002	*Tenrec ecaudatus*	2	2.70	62.1	51.4	11.5	11.0	0.481	13.2	2.73
0003	*Setifer setosus*	1	1.62	41.5	38.3	5.81	5.49	0.316	8.57	1.07
0004	*Hemicentetes semispin.*	1	1.16	19.0	17.1	3.31	3.12	0.188	2.81	0.507
0005	*Echinops telfairi*	1	0.977	17.9	16.4	2.68	2.44	0.242	2.23	0.605
0022	*Erinaceus europaeus*	2	3.71	117	79.9	14.2	13.3	0.903	18.8	3.94
0041	*Sorex minutus*	1	0.161	3.92	3.07	0.707	0.650	0.0569	0.288	0.106
0042	*Sorex araneus*	2	0.264	7.25	5.17	1.09	0.979	0.105	0.610	0.144
0059	*Crocidura occidentalis*	2	0.657	14.6	9.77	1.78	1.63	0.142	2.19	0.252
0060	*Crocidura russula*	1	0.368	7.48	4.61	0.870	0.781	0.0889	0.727	0.116
0063	*Suncus murinus*	1	0.530	12.3	9.67	1.70	1.58	0.116	1.18	0.232

25: Made up of the portion of the internal capsule that passes through the diencephalon.

Table VIII. Volumes of components of the diencephalon (mm³) in primates

Code number	Species	Number of individuals	Epithalamus	Thalamus	Hypothalamus	Subthalamus	Pallidum	Nucleus subthalamicus	Capsula interna	Tractus opticus
			19	20	21	22 (= 23 + 24)	23	24	25	26
0102	*Tupaia glis*	1	1.98	153	47.5	18.6	17.2	1.39	24.6	39.3
0111	*Cheirogaleus major*	1	4.83	271	101	33.8	31.6	2.27	48.6	18.0
0114	*Microcebus murinus*	2	1.44	78.3	29.8	11.4	10.7	0.650	9.08	6.17
0117	*Lepilemur ruficaudatus*	1	4.89	337	114	58.1	55.2	2.91	63.0	20.7
0118	*Lemur fulvus*	1	12.2	1,028	239	225	212	13.9	248	108
0131	*Avahi l. occidentalis*	2	6.52	489	137	80.5	74.5	5.94	105	38.3
0132	*Propithecus verreauxi*	1	14.6	1,341	328	283	263	20.0	324	125
0137	*Loris tardigradus*	1	3.21	348	80.9	45.2	41.4	3.79	46.3	15.7
0139	*Perodicticus potto*	2	10.4	587	160	116	107	8.19	107	19.5
0141	*Galago crassicaudatus*	1	5.71	437	127	67.4	63.3	4.10	74.6	33.9
0142	*Galago demidovii*	2	2.09	153	45.1	18.3	17.0	1.40	18.4	11.5
0147	*Tarsius* sp.	1	2.53	175	42.5	17.1	16.1	1.09	16.7	15.5
0156	*Cebuella pygmaea*	1	3.68	190	69.2	27.7	25.8	1.95	21.8	23.8
0164	*Saguinus oedipus*	2	4.93	410	152	77.6	73.2	4.45	62.7	50.5
0184	*Ateles geoffroyi*	1	30.5	2,930	629	860	807	52.8	570	204
0186	*Lagothrix lagotricha*	2	31.0	2,959	624	909	845	63.8	612	203
0191	*Cebus* sp.	1	23.7	2,163	418	609	567	41.7	549	207
0192	*Saimiri sciureus*	1	8.86	731	207	180	169	11.5	132	117
0209	*Cercocebus albigena*	1	27.9	2,933	681	726	674	52.4	708	308
0212	*Papio anubis*	1	65.4	4,788	1,076	1,241	1,147	94.7	1,257	601
0231	*Cercopithecus ascanius*	1	17.9	1,916	410	431	405	25.7	421	320
0243	*Nasalis larvatus*	1	29.6	3,036	671	587	550	37.2	564	304
0246	*Colobus badius*	2	22.3	2,164	484	553	521	31.9	460	201
0271	*Hylobates lar*	1	31.2	3,226	629	777	715	61.6	693	302
0277	*Pan troglodytes*	1	53.6	7,987	1,739	2,078	1,951	127	2,510	630
0279	*Gorilla gorilla*	1	106	10,547	2,215	2,782	2,604	178	2,471	671
0280	*Homo sapiens sapiens*	1	133	18,222	3,555	5,180	4,711	469	7,008	823

25: Made up of the portion of the internal capsule that passes through the diencephalon.

Table IX. Volumes of visual cortex (area striata) and lateral geniculate body (mm³) in primates

Code number	Species	Number of individuals	Area striata (visual cortex)	Corpus geniculatum laterale	Code number	Species	Number of individuals	Area striata (visual cortex)	Corpus geniculatum laterale
			27	28				27	28
Prosimians					Simians				
0102	*Tupaia glis*	2	167	11.5	0151	*Callithrix jacchus*	1	692	25.7
0106	*Urogale everetti*	1	136	9.98	0164	*Saguinus oedipus*	1	1,003	33.0
0111	*Cheirogaleus major*	2	478	21.5	0165	*Saguinus tamarin*	1	1,077	36.8
0112	*Cheirogaleus medius*	2	217	10.8	0171	*Aotus trivirgatus*	1	1,144	36.0
0114	*Microcebus murinus*	2	149	7.03	0172	*Callicebus moloch*	1	1,502	53.2
0117	*Lepilemur ruficaudatus*	3	357	15.6	0174	*Pithecia monacha*	1	2,154	72.3
0118	*Lemur fulvus*	2	1,527	51.6	0179	*Alouatta* sp.	1	2,374	85.7
0124	*Lemur variegatus*	1	2,112	68.9	0184	*Ateles geoffroyi*	1	4,738	151
0130	*Avahi l. laniger*	1	537	27.4	0186	*Lagothrix lagotricha*	1	6,288	149
0131	*Avahi l. occidentalis*	1	485	27.2	0191	*Cebus* sp.	2	4,690	137
0132	*Propithecus verreauxi*	1	1,559	68.6	0192	*Saimiri sciureus*	1	2,326	62.9
0133	*Indri indri*	1	1,738	99.6	0202	*Macaca mulatta*	1	6,586	158
0136	*Daubentonia madagasc.*	1	1,355	58.5	0209	*Cercocebus albigena*	1	6,831	182
0137	*Loris tardigradus*	1	588	24.6	0227	*Cercopithecus mitis*	1	5,274	150
0138	*Nycticebus coucang*	1	823	37.3	0231	*Cercopithecus ascanius*	1	5,152	147
0139	*Perodicticus potto*	2	552	24.3	0232	*Cercopithecus talapoin*	1	3,046	116
0141	*Galago crassicaudatus*	1	571	22.1	0246	*Colobus badius*	2	3,984	128
0142	*Galago demidovii*	2	237	10.9	0277	*Pan troglodytes*	1	14,691	356
0143	*Galago senegalensis*	1	334	16.0	0279	*Gorilla gorilla*	1	15,185	384
0147	*Tarsius* sp.	2	370	20.8	0280	*Homo sapiens sapiens*	1	22,866	416

27: Included is the underlying white matter.

Table X. Volumes of paleocortex, of amygdala and its various components (mm³) in insectivores

Code number	Species	Number of individuals	Palaeo-cortex	Amygdala	Complexus centro-medialis	Nucleus tractus olfactorii lateralis	Complexus cortico-basolateralis	Nucleus amygdalae basalis, pars magno-cellularis
			29	30 (= 31 + 33)	31	32	33	34
0002	*Tenrec ecaudatus*	1	364	88.1	39.9	3.18	48.1	8.90
0003	*Setifer setosus*	2	269	67.2	30.9	2.71	36.2	7.46
0004	*Hemicentetes semispin.*	1	125	22.3	10.4	1.30	11.9	2.39
0005	*Echinops telfairi*	1	93.1	22.1	10.8	0.710	11.3	3.19
0022	*Erinaceus europaeus*	2	379	104	43.1	3.72	61.1	12.2
0041	*Sorex minutus*	1	13.9	3.34	1.78	0.174	1.56	0.549
0042	*Sorex araneus*	2	25.6	7.19	3.29	–	3.90	0.704
0059	*Crocidura occidentalis*	1	52.7	13.4	6.00	0.403	7.39	1.38
0060	*Crocidura russula*	1	23.4	6.82	3.23	0.233	3.59	0.803
0063	*Suncus murinus*	1	45.1	11.1	5.41	0.440	5.74	0.804

– = Structure was not measured in the species under consideration.

29: Included are retrobulbar cortex (anterior olfactory nucleus), prepiriform cortex, lateral and medial olfactory tracts, internal olfactory tract + anterior commissure, olfactory tubercle, substantia innominata and basal nucleus of Meynert.

31: The centromedial amygdaloid complex is composed of the central and the medial amygdaloid nuclei, and of the anterior amygdaloid area including the nucleus of the lateral olfactory tract. The intercalated cell masses, located at and in part marking the border to the cortico-basolateral amygdaloid complex, were arbitrarily included with the centromedial complex.

33: The cortico-basolateral amygdaloid complex is composed of the lateral, cortical and basal amygdaloid nuclei, the latter with accessory, magnocellular and parvocellular parts.

Table XI. Volumes of paleocortex, of amygdala and its various components (mm^3) in primates; for comments on structures see table X

Code number	Species	Number of individuals	Palaeocortex	Amygdala	Complexus centromedialis	Nucleus tractus olfactorii lateralis	Complexus corticobasolateralis	Nucleus amygdalae basalis, pars magnocellularis
			29	30 (= 31 + 33)	31	32	33	34
0102	*Tupaia glis*	2	182	76.5	24.2	1.71	52.3	4.27
0111	*Cheirogaleus major*	2	233	99.4	34.1	0.851	65.3	7.32
0112	*Cheirogaleus medius*	2	135	63.2	23.3	0.472	39.8	–
0114	*Microcebus murinus*	6	66.8	36.4	12.1	0.264	24.3	–
0117	*Lepilemur ruficaudatus*	3	170	133	42.5	0.624	90.4	7.18
0118	*Lemur fulvus*	3	389	251	71.7	1.37	179	19.6
0132	*Propithecus verreauxi*	1	292	251	78.7	0.921	172	18.8
0136	*Daubentonia madagasc.*	1	1,065	473	122	3.53	351	24.9
0142	*Galago demidovii*	1	122	50.2	14.9	0.620	35.3	2.82
0147	*Tarsius* sp.	1	39.1	49.8	11.1	0.0752	38.7	5.35
0151	*Callithrix jacchus*	3	99.3	113	28.3	0.195	84.6	8.57
0171	*Aotus trivirgatus*	1	203	201	54.5	0.201	147	15.7
0179	*Alouatta* sp.	1	488	413	94.9	0	318	30.7
0191	*Cebus* sp.	1	567	502	118	0	384	41.0
0192	*Saimiri sciureus*	1	186	227	45.9	0	181	15.0
0231	*Cercopithecus ascanius*	1	450	601	136	0	465	48.9
0246	*Colobus badius*	2	458	479	114	0	365	45.5
0280	*Homo sapiens sapiens*	1	6,016	3,015	592	0	2,424	246

Table XII. Volumes of vestibular complex and its various components (mm³) in insectivores

Code number	Species	Number of individuals	Complexus vestibularis	Nucleus vestibularis superior	Nucleus vestibularis lateralis	Nucleus vestibularis medialis	Nucleus vestibularis descendens
			35 (= 36–39)	36	37	38	39
0002	*Tenrec ecaudatus*	2	7.59	1.23	1.33	2.68	2.35
0003	*Setifer setosus*	1	3.75	0.689	0.483	1.44	1.14
0004	*Hemicentetes semispin.*	1	2.50	0.384	0.552	0.852	0.709
0005	*Echinops telfairi*	1	1.69	0.291	0.281	0.615	0.505
0022	*Erinaceus europaeus*	1	7.54	1.21	1.72	2.85	1.77
0041	*Sorex minutus*	1	0.410	0.0482	0.0713	0.174	0.116
0042	*Sorex araneus*	1	0.618	0.107	0.0955	0.249	0.166
0059	*Crocidura occidentalis*	2	1.32	0.171	0.222	0.543	0.389
0060	*Crocidura russula*	1	0.563	0.0770	0.0840	0.235	0.167
0063	*Suncus murinus*	1	1.03	0.100	0.167	0.438	0.324

calculated as percent of the mean. The 720 SEM percent values (48 × 15) vary from 0 to 88.9. The degree of variability is obviously not the same in all structures. In some structures, high values predominate, in others, low values. When averaging the SEM values, the highest variability was found in a complex comprised of meninges, nerves, etc. (average 28.5, max. 88.9). The main components are the stumps of the optic nerves until they converge in the chiasma, the trigeminal nerve with the Gasserian ganglion (which, in our preparations, is always left in place unilaterally), the hypophysis, and the remnants of the meninges. The hypophysis is included in this complex to obtain comparable values for the diencephalon and for the net brain tissue of the total brain. The hypophysis is sometimes broken (especially in human brains) and then not available for measurements.

The complex of meninges, nerves, etc. is included in the measurements even though its constituent parts are not, strictly speaking, brain tissue. Its inclusion is appropriate since its parts are also included when the whole brain is weighed. The complex is relatively small and amounts only to about 1% of the total brain. Omitting it would not greatly distort the picture obtained of the composition of the brain. Its high variability is mainly due to minor differences in the brain preparation.

A similar high variability is obtained for the ventricles, another complex that does not consist of true brain tissue (average 21.9, max. 51.6). The ventricles were included in our measurements because they are expected to be filled with liquid when weighing the brain and thus to influence the brain weight. During embedding which gives rise to considerable shrinkage, the ventricles may behave differently in individual brains, thereby resulting in the relatively high variability shown here.

Table XIII. Volumes of vestibular complex and its various components (mm^3) in primates

Code number	Species	Number of individuals	Complexus vestibularis	Nucleus vestibularis superior	Nucleus vestibularis lateralis	Nucleus vestibularis medialis	Nucleus vestibularis descendens
			35 (= 36–39)	36	37	38	39
0102	*Tupaia glis*	1	10.1	2.16	2.07	3.27	2.62
0111	*Cheirogaleus major*	1	11.1	1.86	1.78	4.18	3.27
0114	*Microcebus murinus*	2	4.02	0.679	0.760	1.50	1.08
0117	*Lepilemur ruficaudatus*	1	14.9	2.11	2.38	5.92	4.47
0118	*Lemur fulvus*	1	31.1	6.22	5.90	9.75	9.27
0131	*Avahi l. occidentalis*	2	18.1	3.10	3.32	6.86	4.81
0132	*Propithecus verreauxi*	1	43.2	8.18	7.37	15.3	12.4
0137	*Loris tardigradus*	1	8.45	1.61	1.22	3.55	2.08
0139	*Perodicticus potto*	2	17.3	2.86	2.66	7.87	3.88
0141	*Galago crassicaudatus*	1	19.9	2.85	2.87	7.16	7.01
0142	*Galago demidovii*	2	6.65	1.02	1.01	2.72	1.90
0147	*Tarsius* sp.	1	7.69	0.759	1.16	3.36	2.41
0156	*Cebuella pygmaea*	1	11.1	2.10	2.94	4.11	1.96
0164	*Saguinus oedipus*	2	18.4	2.98	3.63	7.25	4.49
0184	*Ateles geoffroyi*	1	49.7	7.20	6.81	23.8	11.9
0186	*Lagothrix lagotricha*	2	53.0	9.07	6.94	23.0	14.1
0191	*Cebus* sp.	1	50.2	11.2	7.45	19.2	12.3
0192	*Saimiri sciureus*	1	29.4	4.43	6.26	11.4	7.31
0209	*Cercocebus albigena*	1	87.1	16.2	10.0	35.6	25.2
0212	*Papio anubis*	1	107	14.4	10.5	48.1	33.9
0231	*Cercopithecus ascanius*	1	52.0	8.21	5.88	20.8	17.2
0243	*Nasalis larvatus*	1	86.2	12.6	14.8	33.6	25.2
0246	*Colobus badius*	2	57.3	8.68	10.0	24.1	14.5
0271	*Hylobates lar*	1	59.3	14.9	5.53	24.5	14.5
0277	*Pan troglodytes*	1	118	23.5	9.65	54.7	29.8
0279	*Gorilla gorilla*	1	136	34.7	19.5	58.4	23.6
0280	*Homo sapiens sapiens*	1	177	35.8	22.4	72.8	45.9

The highest variability of a structure consisting of true brain tissue is found in the olfactory bulbs (main and accessory bulbs). The accessory olfactory bulb has an average SEM of 18.0% with a maximum of 82.0%, the main bulb an average of 10.9% and a maximum of 26.8%. In a descending sequence are the schizocortex (average 8.8, max. 21.3), septum (8.4, 30.5, respectively), medulla oblongata (8.1, 25.1), piriform lobe (palaeocortex + amygdala; 7.6, 25.7), mesencephalon (7.1, 19.6), cerebellum (6.9, 19.2),

Table XIV. Volumes of various periventricular structures (mm^3) in insectivores

Code number	Species	Number of individuals	Nucleus septalis triangularis	Corpus subfornicale	Nucleus habenularis medialis	Corpus pineale	Corpus subcommissurale
			40	41	42	43	44
0001	*Solenodon paradoxus*	1	0.870	0.0830	2.08	1.27	0.0435
0002	*Tenrec ecaudatus*	1	0.704	0.124	0.988	0.480	0.0112
0003	*Setifer setosus*	1	0.329	0.0600	0.426	0.445	0.00310
0004	*Hemicentetes semispin.*	1	0.338	0.0320	0.425	0.181	0.00382
0005	*Echinops telfairi*	1	0.338	0.0290	0.365	0.243	0.000212
0006	*Oryzorictes talpoides*	1	0.331	0.0180	0.257	0.0510	0.000694
0007	*Microgale cowani*	1	0.216	0.0116	0.147	0.0152	0.00129
0008	*Limnogale mergulus*	1	0.602	0.0194	0.469	0.122	0.00202
0009	*Nesogale dobsoni*	1	0.280	0.00760	0.242	0.0356	0.000950
0010	*Nesogale talazaci*	1	0.378	0.0191	0.381	0.0894	0.00147
0015	*Potamogale velox*	1	1.10	0.0890	1.45	0.00309	0.00174
0016	*Chlorotalpa stuhlmanni*	6	0.395	0.0191	0.380	0.0960	0
0017	*Chrysochloris asiatica*	1	0.485	0.0147	0.449	0.0251	0.00811
0021	*Aethechinus algirus*	1	1.18	0.0621	0.930	0.297	0.0222
0022	*Erinaceus europaeus*	2	0.887	0.0592	1.17	0.423	0.0312
0031	*Elephantulus fuscipes*	6	0.289	0.0709	0.415	0.336	0.124
0033	*Rhynchocyon stuhlmanni*	1	0.937	0.0576	2.09	1.41	0.403
0041	*Sorex minutus*	1	0.0700	0.00284	0.0586	0.00359	0.000265
0042	*Sorex araneus*	10	0.115	0.00326	0.0749	0.0102	0.000120
0043	*Neomys fodiens*	1	0.263	0.00690	0.205	0.0270	0.000418
0059	*Crocidura occidentalis*	1	0.274	0.0186	0.200	0.0391	0.000576
0060	*Crocidura russula*	5	0.120	0.00986	0.134	0.0278	0.000766
0063	*Suncus murinus*	1	0.201	0.0148	0.168	0.0628	0.00223
0081	*Talpa europaea*	5	0.330	0.00857	0.366	0.0497	0.0102
0083	*Desmana moschata*	1	1.06	0.0352	1.05	0.459	0.0278
0084	*Galemys pyrenaicus*	5	0.444	0.0379	0.422	0.0707	0.00545

striatum (6.9, 20.3), hippocampus (5.5, 15.0), diencephalon (5.2, 13.6), neocortex (3.4, 16.7), and telencephalon (3.1, 6.5). The differences in variability of the various structures seem to be mainly due to true intraspecific variation and only to a small degree to difficulties in delineation. Structures that are difficult to delineate such as diencephalon, striatum, and mesencephalon have the same or even lower standard deviations than those that are easy to delineate (olfactory bulbs, cerebellum).

In *Tupaia* 12 main structures (without meninges, ventricles and accessory olfactory bulbs) in 20 individuals were compared and the individual deviation from the mean was determined for each structure. The percentage

Table XV. Volumes of various periventricular structures (mm^3) in prosimians

Code number	Species	Number of individuals	Nucleus septalis triangularis	Corpus subfornicale	Nucleus habenularis medialis	Corpus pineale	Corpus subcommissurale
			40	41	42	43	44
0102	*Tupaia glis*	6	0.454	0.0608	0.947	0.393	0.0804
0106	*Urogale everetti*	2	0.660	0.136	0.926	1.42	0.0572
0111	*Cheirogaleus major*	1	0.917	0.256	1.75	0.789	0.0320
0112	*Cheirogaleus medius*	1	0.385	0.171	1.06	0.691	0.0432
0114	*Microcebus murinus*	1	0.274	0.0556	0.461	0.140	0.0233
0117	*Lepilemur ruficaudatus*	1	0.888	0.121	1.75	0.731	0.132
0118	*Lemur fulvus*	4	1.40	0.527	4.02	1.85	0.174
0124	*Lemur variegatus*	1	3.51	0.741	5.35	4.46	0.197
0130	*Avahi l. laniger*	1	1.29	0.116	2.21	0.700	0.140
0131	*Avahi l. occidentalis*	1	0.988	0.188	2.04	0.652	0.0942
0132	*Propithecus verreauxi*	1	1.60	0.507	3.65	3.26	0.357
0133	*Indri indri*	1	4.96	0.553	7.15	5.63	0.366
0136	*Daubentonia madagasc.*	1	2.65	0.593	7.75	3.32	1.83
0137	*Loris tardigradus*	1	0.822	0.207	1.15	0.335	0.103
0138	*Nycticebus coucang*	1	1.34	0.175	2.04	1.83	0.154
0139	*Perodicticus potto*	5	2.27	0.216	3.77	1.76	0.264
0141	*Galago crassicaudatus*	1	0.695	0.0948	1.64	1.02	0.102
0142	*Galago demidovii*	2	0.450	0.0240	0.780	0.375	0.0266
0143	*Galago senegalensis*	1	0.588	0.0668	0.998	0.741	0.0382
0147	*Tarsius* sp.	3	0.405	0.0482	1.13	2.03	0.114

average of the total of all 240 values (12 structures of 20 individuals) was found to be 3.9 ranging from 0.1 to 16.9. The maximal deviation of 16.9 again was found in the olfactory bulb. Of all 240 values, 171 (71%) fell below 5, and 226 (94%) below 10. These results confirm those obtained from more restricted material [Stephan et al., 1970] as well as those found in Chiroptera [Stephan and Pirlot, 1970].

In random samples, a deviation of less than 10% from the mean for all 20 specimens was nearly always reached when investigating 2 individuals. In the most unfavorable sequence of the *Tupaia* specimens which is theoretically possible, a maximum of 7 brains was necessary for measurements

Table XVI. Volumes of various periventricular structures (mm^3) in simians

Code number	Species	Number of individuals	Nucleus septalis triangularis	Corpus subfornicale	Nucleus habenularis medialis	Corpus pineale	Corpus subcommissurale
			40	41	42	43	44
0151	*Callithrix jacchus*	1	0.440	0.133	1.63	0.486	0.0789
0164	*Saguinus oedipus*	1	1.08	0.167	1.55	0.392	0.0921
0165	*Saguinus tamarin*	1	0.559	0.0795	1.18	0.216	0.0994
0171	*Aotus trivirgatus*	1	0.964	0.162	3.21	1.55	0.140
0172	*Callicebus moloch*	1	0.890	0.135	2.66	2.90	0.154
0174	*Pithecia monacha*	1	1.31	0.300	2.53	0.0623	0.189
0179	*Alouatta* sp.	1	1.72	0.372	3.30	0.254	0.402
0184	*Ateles geoffroyi*	2	3.18	0.443	7.29	3.10	0.608
0186	*Lagothrix lagotricha*	2	2.60	0.583	7.01	10.6	0.885
0191	*Cebus* sp.	2	1.50	0.410	4.90	3.80	0.425
0192	*Saimiri sciureus*	5	0.598	0.255	2.77	2.71	0.143
0202	*Macaca mulatta*	1	2.83	0.360	7.66	12.4	0.270
0209	*Cercocebus albigena*	1	1.69	0.658	6.00	11.9	0.233
0212	*Papio anubis*	1	1.86	0.405	12.7	51.1	0.347
0227	*Cercopithecus mitis*	1	1.96	0.432	5.38	15.3	0.258
0231	*Cercopithecus ascan.*	1	2.13	0.509	4.23	4.66	0.217
0232	*Cercopithecus talap.*	3	0.866	0.192	2.07	2.03	0.220
0246	*Colobus badius*	1	2.26	0.454	4.54	7.31	0.265
0277	*Pan troglodytes*	3	2.70	0.606	9.40	38.0	0.387
0279	*Gorilla gorilla*	2	5.79	1.30	23.9	7.96	1.41
0280	*Homo sapiens sapiens*	1	0.524	0.508	17.0	142	0.0203

of the olfactory bulb to arrive at an average deviation lower than 10%, 5 for the medulla oblongata and 4 for the hippocampus. However, such sequences are extremely unlikely. In 9 of the 12 structures, an average deviation lower than 10% was reached even under the most unfavorable sequences when investigating 2 brains.

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